140 patients with CLBP of at least 6 months duration, with Modic Type 1 or 2 vertebral endplate changes between L3 and S1, were randomized 1:1 to undergo either Radiofrequency (RF) ablation of the BVN or continue standard care. Oswestry Disability Index (ODI) was collected at baseline, 3, 6, 9, and 12-months post procedure. Secondary outcome measures included a 10-point Visual Analog Scale (VAS) for LBP, ODI and VAS responder rates, SF-36, and EQ-5D-5L. The primary endpoint was a between-arm comparison of the mean change in ODI from baseline to 3 months post-treatment.

The interim analysis showed clear statistical superiority (p<.001) for all primary and secondary patient-reported outcome measures in the RF ablation arm compared with the standard care arm. This resulted in a data management committee recommendation to halt enrollment in the study and offer early cross-over to the control arm. These results are comprised of the outcomes of the 104 patients included in the intent-to-treat analysis of the 3-month primary endpoint, which included 51 patients in the RF ablation arm and 53 patients in the standard care arm. Baseline ODI was 46.1, VAS was 6.67, and mean age was 50 years. The percentage of patients with LBP symptoms ≥5 years was 67.3%. Comparing the RF ablation arm to the standard care arm, the mean changes in ODI at 3 months were −25.3 points versus −4.4 points, respectively, resulting in an adjusted difference of 20.9 points (p<.001). Mean changes in VAS were −3.46 versus −1.02, respectively, an adjusted difference of 2.44 cm (p<.001). In the RF ablation arm, 74.5% of patients achieved a ≥10-point improvement in ODI, compared with 32.7% in the standard care arm (p<0.001).

CONCLUSIONS: Minimally invasive RF ablation of the BVN led to significant improvement of pain and function at 3-months in patients with chronic vertebrogenic related LBP.

A Prospective, randomized, multicenter study of intraosseous basivertebral nerve ablation for the treatment of chronic low back pain: 24-Month treatment arm results was published October 2021 (Koreckij et.al). Results: 140 patients were randomized, 66 to BVNA. In the 58 BVNA patients completing a 24-month visit, 67% had back pain for >5 years, 36% were actively taking opioids at baseline, 50% had prior epidural steroid injections, and 12% had prior low back surgery. Improvements in ODI, VAS, SF-36 PCS, and EQ-5D-5L were statistically significant at all timepoints through 2 years. At 24 months, ODI and VAS improved 28.5±16.2 points (from baseline 44.5; p < 0.001) and 4.1±2.7 cm (from baseline 6.6; p < 0.001), respectively. A combined responder rate of ODI≥15 and VAS≥2 was 73.7%. A ≥50% reduction in pain was reported in 72.4% of patients and 31.0% were pain-free at 2 years. At 24 months, only 3(5%) of patients had BVNA-level steroid injections, and 62% fewer patients were actively taking opioids. There were no serious device or device-procedure related adverse events reported through 24 months.

In conclusion, Intraosseous BVNA demonstrates an excellent safety profile and significant improvements in pain, function, and quality of life that are sustained through 24 months in patients with chronic vertebrogenic low back pain.